

Networking Essentials

Whenever we want to access a network we need to configure the following:

1. IP address with Subnet mask.
2. Router (Or gateway).
3. DNS Server.

We can either do this manually or automatically saving us time using a DHCP -> Dynamic Host Configuration Protocol.

It is an application level protocol that relies on UDP.

DHCP follows 4 steps:

1. **DHCP Discover: DHCPDISCOVER.**
2. **DHCP Offer: DHCPOFFER.**
3. **DHCP Request: DHCPREQUEST.**
4. **DHCP Acknowledge: DHCPACK.**

This process is known as “DORA”.

ICMP - Is a great tool for troubleshooting, error reporting and network diagnostics. We can use this in one of two ways:

Ping: Uses ICMP to test connectivity to a target system. This also measures a round-trip time (RTT) in ms.

Traceroute: Uses ICMP to discover the route taken from host to destination.

When we want to communicate with other networks such as those which require a gateway, there are some different algorithms that are used:

OSPF (Open Shortest Path First): Share information about the network topology and determine the most efficient / shortest path for transmission.

EIGRP (Enhanced Interior Gateway Routing Protocol): Cisco proprietary routing protocol which combines factors such as information about the networks they can reach and the cost (bandwidth or delay) associated with those routes.

BGP (Border Gateway Protocol): Primary routing used on the internet. Allows different networks to exchange routing information and establish paths for data to travel between these networks. Can be routed efficiently.

RIP (Routing Information Protocol): Is a simple routing protocol often used in small networks. Routers running RIP share information about the networks they can reach and the number of hops (routers) required to get there.

ARP: Address resolution Protocol, establish the identity of a device on a network.